HeatShield AI – Prototype Build Guide (Google AI Studio + ESP32)

Team Tech Forge · Future Health Digital Innovation Challenge 2026 · Climate-Resilient Worker Health

1. What you are building

A working web prototype with three views (Worker, Rider, Supervisor), a rule-first heat-risk engine, a Gemini reasoning layer (explanations, SmartShift plan, multilingual alerts) and a live ESP32 sensor feed with a labelled simulated fallback. Effort split: about 80% software, 20% hardware. Guest access only, no login, so judges can open it instantly.

Build order (do not skip ahead): 1) app with simulated sensor, 2) Gemini layer, 3) ESP32 live feed, 4) demo video and polish.

2. Architecture and data flow

- **ESP32 node** reads a DHT22 (air temperature, humidity) and a DS18B20 in a matte-black sphere (globe temperature) every 5 seconds.
- The ESP32 sends a JSON reading by HTTPS to a **Firebase Realtime Database** (REST). It is the simplest relay, because a browser app cannot reach an ESP32 on a local network (HTTPS app calling an HTTP device is blocked).
- The **AI Studio web app** polls that database URL every 3 seconds. If it gets nothing, it switches to **Simulated sensor** mode and shows a visible "SIMULATED" badge.
- **Rule engine** (plain JavaScript) computes WBGT estimate and the risk level. **Gemini** only explains, plans and translates; it can raise a level but never lower it.

Reading format the app expects:

```
{ "t": 38.4, "rh": 52, "globe": 49.1, "ts": 1760000000000 }
```

3. Risk engine (illustrative, not medically validated)

All thresholds below are placeholders for the demo. Say so on screen, and replace them with values from ISO 7243, ACGIH or Dubai Municipality guidance after you verify them.

Step 1: wet-bulb estimate (Stull 2011). T in °C, RH in %:

```
const wetBulb = (T, RH) =>
  T*Math.atan(0.151977*Math.sqrt(RH+8.313659)) + Math.atan(T+RH)
  - Math.atan(RH-1.676331) + 0.00391838*Math.pow(RH,1.5)*Math.atan(0.023101*RH) -
4.686035;
```

Step 2: WBGT estimate. With a globe probe: `WBGT = 0.7*Tnwb + 0.2*Tg + 0.1*Ta`. Without it, use a demo sun selector that adds an assumed offset to Tg (shade +0, partial sun +3, full sun +7 °C) and label it as an assumption.

Step 3: margin against a workload threshold (illustrative).

Workload	Placeholder WBGT threshold (°C)

Workload	Placeholder WBGT threshold (°C)
Light	31
Moderate	28
Heavy	26
Very heavy	24

Subtract 3 °C from the threshold if the worker is not acclimatized, 2 °C for heavy PPE or clothing, 1 °C if exposure exceeds 4 hours.

Step 4: level from margin = threshold − WBGT.

Margin	Level	Safe action
3 or more	Low	Normal work, hydrate
0 to 3	Moderate	Hydrate, monitor, shade breaks
−3 to 0	High	Cooling break, review workload
below −3	Critical	Stop work, urgent safety protocol

Overrides: any reported symptom means at least High. Between 12:30 and 15:00 from 15 June to 15 September, the status is "Outdoor work not permitted (UAE midday ban)" regardless of the score. Never suggest bypassing it.

4. Paste this prompt into Google AI Studio (Build tab)

Open Google AI Studio, choose the Build option for creating an app, and paste the prompt below. It is written to be self-contained. After the first version appears, iterate with the follow-up prompts in section 5.

```
Build a responsive web app called "HeatShield AI": a privacy-first heat-risk early-warning and SmartShift planning tool for outdoor workers in Dubai/UAE. It is decision support, NOT a medical device; show this note in the footer.

ACCESS: No login. The first screen has a large "Continue as Guest" button, then a role picker: Worker, Delivery Rider, Supervisor. A role switcher stays in the top bar. All features work for guests.

DATA SOURCES: (a) Live mode: poll a configurable URL (input field, default empty) every 3 seconds expecting JSON {t, rh, globe, ts}. (b) Simulated mode (default when the URL is empty or fails): sliders for air temperature (25-50 C), humidity (20-90 %), sun exposure (shade/partial/full), plus a button "Run heat-wave scenario" that ramps conditions over 60 seconds. Always show a badge LIVE or SIMULATED.

RISK ENGINE (plain JavaScript, deterministic, shown in a "How this score works" panel): estimate wet-bulb with the Stull formula, then WBGT = 0.7*wetBulb + 0.2*globe + 0.1*air (if no globe reading, globe = air + 0/3/7 for shade/partial/full sun). Workload selector: light 31, moderate 28, heavy 26, very heavy 24 (placeholder thresholds, label as illustrative). Subtract 3 if not acclimatized, 2 if heavy PPE, 1
```

```
if exposure over 4 hours. margin = threshold - WBGT. Level: margin>=3 Low (green), 0-3 Moderate (amber), -3-0 High (orange), < -3 Critical (red). Any reported symptom forces at least High. Between 12:30 and 15:00 from June 15 to Sept 15 show "Outdoor work not permitted: UAE midday ban" (add a demo toggle to simulate this time). The AI layer may raise a level but never lower it.

GEMINI LAYER (use the Gemini API available in AI Studio; request JSON-only responses with a response schema): call only when the level changes or the user presses a button, not every poll. Provide inputs as structured data (no names). Return: {explanation (2 sentences naming the top 3 drivers), recommended_actions[3], level_suggestion, alert_text per language}. Languages: English, Hindi, Malayalam, Kannada, Arabic, Urdu. Keep alerts under 25 words, simple words, one safe action. Example: "Heat risk is increasing. Please pause work, move to a shaded or cooled area, and inform your supervisor." Add a speaker button that reads the alert aloud with the browser speechSynthesis API in the chosen language when a voice exists.

WORKER VIEW: big risk card (color + word + icon, not color only), current WBGT estimate, drivers, the multilingual alert with language dropdown, hydration/rest guidance, a voluntary symptom check-in (checkboxes: dizziness, headache, nausea, cramps, none) with the note "Optional and private", and a red SOS button that shows a mock escalation message to the supervisor.

RIDER VIEW: same risk card plus "Nearest cooling stations": use a demo dataset of 6 stations around Dubai (clearly labelled DEMO DATA, with name, lat, lng, and an "availability" shown only as "unknown (demo)"). Ask for the browser location, fall back to a default Dubai location, sort by haversine distance, show distance and walking/riding minutes, and a simple map. When risk is High or Critical, pin the nearest station at the top with a "Go now" button that opens a maps directions link.

SUPERVISOR VIEW: a table of 3 demo crews (Crew A moderate workload, Crew B heavy, Crew C very heavy) with workload, exposure hours, risk level, suggested action. NO individual names or medical details, crew-level only; show a privacy banner. A "SmartShift Planner" button asks Gemini for a plan: which heavy tasks to move to cooler hours, break timing, rotation between crews, which tasks need review, while keeping within UAE rules and never bypassing the midday ban. Show the plan as a timeline plus a short rationale. Include a small trend chart of WBGT over the last 5 minutes and an "Alert feedback" button (useful / false alarm) that updates counters for alert lead time and false-alarm rate.

TRUST FEATURES: every alert has an "Why this alert?" drawer listing the inputs and the rule that fired. A footer notes thresholds are illustrative, data is synthetic, and no personal data is stored. Use a clean, high-contrast, mobile-first design; large touch targets; respect prefers-reduced-motion.

ERROR HANDLING: if the sensor URL fails, switch to simulated mode with a visible banner; if Gemini fails, fall back to template text for English and keep the rule-based alert working.
```

5. Follow-up prompts to paste one at a time

1. "Add a Demo Controls drawer with: sensor URL input, LIVE/SIMULATED switch, midday-ban simulation toggle, and a reset button."
2. "Make the Supervisor view show a 3-crew SmartShift timeline from 06:00 to 18:00 with the 12:30-15:00 block greyed out as the legal midday ban."

3. "Add an 'Impact' tab showing alert lead time, false-alarm rate, and adherence counters from the feedback buttons, with a note that targets are pilot hypotheses."
4. "Review the app for accessibility: add aria-labels, ensure risk is never conveyed by color alone, and check contrast."
5. "Add a small 'Evidence' panel citing Ahmed et al. 2020 (Industrial Health 58(2):170-181) and Bates and Schneider 2008 (J Occup Med Toxicol 3:21)."

6. ESP32 hardware setup

Parts: ESP32 DevKit, DHT22 (or SHT31), DS18B20 probe in a matte-black sphere (improvised globe), 4.7 kΩ pull-up resistor for the DS18B20, breadboard and jumper wires, USB power, optional 0.96-inch OLED.

Wiring: DHT22 data to GPIO 4 (3.3 V and GND); DS18B20 data to GPIO 5 with the 4.7 kΩ resistor between data and 3.3 V.

Firebase relay (about 10 minutes): create a Firebase project, add a Realtime Database, and for the demo only set rules to `{ "rules": { ".read": true, ".write": true } }`. Delete or lock the database after the event. It holds no personal data. Copy the database URL.

Arduino IDE libraries: DHT sensor library (Adafruit), OneWire, DallasTemperature. Board package: esp32 by Espressif.

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <WiFiClientSecure.h>
#include <DHT.h>
#include <OneWire.h>
#include <DallasTemperature.h>

#define DHTPIN 4
#define DHTTYPE DHT22
#define ONEWIRE_PIN 5

const char* WIFI_SSID = "YOUR_WIFI";
const char* WIFI_PASS = "YOUR_PASSWORD";
const char* DB_URL = "https://YOUR-PROJECT-default-
rtdb.firebaseio.com/heatshield/node1.json";

DHT dht(DHTPIN, DHTTYPE);
OneWire ow(ONEWIRE_PIN);
DallasTemperature globe(&ow);

void setup() {
  Serial.begin(115200);
  dht.begin();
  globe.begin();
  WiFi.begin(WIFI_SSID, WIFI_PASS);
  while (WiFi.status() != WL_CONNECTED) { delay(500); Serial.print("."); }
  Serial.println("\nWiFi connected");
}

void loop() {
```

```
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  globe.requestTemperatures();
  float g = globe.getTempCByIndex(0);
  if (isnan(t) || isnan(h)) { delay(2000); return; }
  String gs = (g == DEVICE_DISCONNECTED_C) ? "null" : String(g, 1);
  String body = "{\"t\":" + String(t, 1) + ",\"rh\":" + String(h, 0) +
                ",\"globe\":" + gs + ",\"ts\":{\".sv\":\"timestamp\"}}";
  WiFiClientSecure client;
  client.setInsecure();            // demo only; use a pinned certificate in a pilot
  HTTPClient http;
  http.begin(client, DB_URL);
  http.addHeader("Content-Type", "application/json");
  int code = http.PUT(body);
  Serial.printf("HTTP %d  %s\n", code, body.c_str());
  http.end();
  delay(5000);
}
```

Connect to the app: in the web app's Demo Controls, paste `https://YOUR-PROJECT-default-rtdb.firebaseio.com/heatshield/node1.json` as the sensor URL. The badge should flip to LIVE. If the app cannot read it, ask AI Studio: "Fix CORS or fetch errors when polling the sensor URL and show the error in the banner."

Live demo trick: hold a hair dryer or warm hand near the DHT22 (from a safe distance, never heating the board) so the reading visibly climbs while the risk level rises on screen.

7. Test checklist before you record

- Simulated mode: sliders change WBGT and the level within 1 second.
- Heat-wave scenario reaches Critical and the alert is spoken in at least English, Hindi and Kannada.
- Midday-ban toggle overrides the score with the legal message.
- Symptom check-in forces at least High.
- Unplug the ESP32: the app falls back to SIMULATED with a banner, nothing crashes.
- Gemini turned off or failing: template alert still appears.
- Supervisor view shows crews only, with no individual names or medical details.
- Open the published link in a private window and confirm Continue as Guest works.

8. Publish and wire into the PPT

1. In AI Studio use the share or deploy option available for built apps to get a public link, then open it in a private window to confirm guest access.
2. Paste the link on PPT slide 11 (login/guest) and slide 17 (demo flow).
3. Record the 3-minute demo following slide 17 (worker, live sensor, risk rise, multilingual alert, supervisor SmartShift plan, rider cooling station, impact).
4. Replace the placeholders in the PPT: teammate name, organization, 18+ confirmation for each member, conflict-of-interest confirmation.

9. Honesty guardrails for the pitch

- Say "illustrative thresholds" and "decision support, not a medical device".
- Do not claim a number of Dubai workers with heat illness; no reliable citywide figure exists in your research.
- Do not claim the sensors predict heat stroke. They estimate environmental heat load.
- Label any station availability as demo unless a real data source is connected.
- Evidence note: Bates and Schneider (2008) found WBGT alone can misstate risk in Gulf conditions, so present your approach as workload-aware and personalised rather than WBGT-only.

10. Suggested time plan

Your portal screenshot showed about two days left when it was captured, so check the live countdown first.

- **First half:** AI Studio app with simulated mode, rule engine and the three views.
- **Second half:** Gemini layer, multilingual alerts, SmartShift planner.
- **Then:** ESP32 firmware and Firebase relay, test checklist, record video.
- **Last:** fill the PPT placeholders, add the link, submit early.

11. Research: existing solutions and their gaps

This is a review of public sources, not an exhaustive market survey. Numbers in brackets point to the references in section 13.

Existing solution	What it does	Gap HeatShield AI addresses
UAE MoHRE midday break and rest stations [4][5]	Midday ban 12:30–3:00 PM from 15 June to 15 September (22nd year in 2026), shaded breaks, inspections. 6,000 delivery rest stations with an interactive map in 2024; 10,000+ announced in 2025	One fixed rule for everyone. No live risk by crew, task or weather. The station map is not risk- or route-aware
Kenzen wearable, trialled by EGA in the UAE [6][7]	Arm patch estimates core temperature and alerts worker and supervisor. Sold per worker per month. EGA's 2023 trial gave it to 350 volunteers	Needs a body-worn device on every worker, so higher cost and a bigger privacy burden. Built around individuals, not crew scheduling or roaming riders
OSHA-NIOSH Heat Safety Tool [3][8]	Free app: heat index and hourly forecast from local weather, precautions, reminders	Weather-only. A 2020 field comparison found it less reliable as risk rose, versus a WBGT monitor. US-oriented. No crew, workload or scheduling
Certified WBGT meters [3]	Reference measurement of site heat stress	Accurate but manual and point-in-time. People must read and act. No multilingual

Existing solution	What it does	Gap HeatShield AI addresses
		alerts or planning
Cooling aids: cool collars, AC safety helmet [9]	Reduce heat load physically	No data or early warning. Complementary to HeatShield AI

Scientific context. A UAE construction-site study found WBGT above the recommended limit and medium-to-high heat-stress risk, and advised rescheduling work and shaded breaks [1]. An earlier UAE study found that WBGT alone can misstate risk in Gulf conditions because workers self-pace and hydrate, which argues for workload-aware, personalised assessment [2].

What the gap looks like. Existing tools mostly monitor and alert. In our review we found none that combines task-aware scheduling, predictive heat windows and rider cooling routes without requiring a body-worn device. Say "in our review" in the pitch, not "no one has done this".

12. Innovation: what HeatShield AI adds, and how to build each in AI Studio

#	Innovation	Why it is new versus the table above	Build it (prompt to paste into AI Studio)
1	**Sensor-light, wearable-optional risk**	Works from a low-cost site node plus forecast; body-worn data is an optional consented extra. Lower cost and privacy risk than per-worker wearables	"Add an optional 'Wearable data' toggle (off by default). When on, accept a demo heart-rate value and show it only as a crew-level aggregate in the Supervisor view."
2	**Predictive Heat Windows**	Looks ahead instead of only alerting now	"Fetch an hourly forecast (temperature and humidity) from the Open-Meteo API for Dubai, compute WBGT for the next 6 hours with the same rule engine, and show a bar of safe, caution and stop windows. Mark 12:30–15:00 as the legal midday ban."
3	**SmartShift Planner using the windows**	Moves heavy tasks into cooler windows, task-aware, never bypassing the ban	"Let Gemini take the crews, tasks and heat windows and return a JSON schedule that puts heavy tasks in the lowest-WBGT windows, adds rotation and breaks, and never schedules outdoor work in the midday ban."
4	**Micro-zone calibration**	Site node versus forecast gives a site-specific offset, so	"Keep a rolling average of (sensor WBGT minus forecast WBGT) over the last 30 minutes, show it as the 'site offset', and apply it to the forecast

#	Innovation	Why it is new versus the table above	Build it (prompt to paste into AI Studio)
		alerts match the real worksite (sun versus shade)	windows. Show it as a small hot-zone map for 3 demo zones."
5	**Cool-by routing for riders**	Station suggestion is tied to when the rider will cross the High band	"In Rider view, estimate minutes until the High band from the current WBGT trend, then list only cooling stations reachable within that time, nearest first, with a 'Cool by HH:MM' label."
6	**Worker data ledger**	Worker-first transparency: workers see exactly what is shared and with whom	"Add a 'My data' screen in Worker view listing every data item collected, who can see it (worker only, crew-level, none), and a button to turn off optional items."
7	**Compliance evidence pack**	Anonymised, inspection-ready log of breaks and ban compliance	"Add a Supervisor 'Compliance report' that exports a CSV of anonymised crew-level events: alert time, level, action taken, break logged, midday-ban adherence."
8	**Acclimatization ramp and buddy check-ins**	Protects new or returning workers and avoids working alone	"Add a 'New or returning worker' flag that lowers the threshold and shows a gradual exposure schedule. Add a buddy check prompt every 30 minutes when risk is Moderate or higher."
9	**Feedback-tuned thresholds with a safety floor**	Reduces alert fatigue without weakening safety	"Use the 'false alarm' feedback to suggest a threshold change, but never allow thresholds to move below the configured safety floor. Show every change in a log."

Pitch line. "HeatShield AI does not replace the midday ban, wearables or cooling aids. It adds the missing planning layer: predict the heat, schedule around it, route to cooling in time, and keep worker data private."

Evidence you can show for each claim. Innovations 2 to 4 rest on the WBGT approach and Gulf-specific findings [1][2][3]. Innovation 1 responds to the cost and privacy burden of per-worker wearables [6]. Say clearly that the thresholds and the forecast-window logic are untested prototypes, and that your pilot plan is how you will validate them.

13. References

1. Ahmed HO et al. Assessment of thermal exposure level among construction workers in UAE using WBGT, HSI and TWL indices. Industrial Health 2020;58(2):170–181. https://pubmed.ncbi.nlm.nih.gov/31308288/

2. Bates GP, Schneider J. Hydration status and physiological workload of UAE construction workers. Journal of Occupational Medicine and Toxicology 2008;3:21. https://link.springer.com/content/pdf/10.1186/1745-6673-3-21.pdf
3. Dillane D, Balanay JG. Comparison between OSHA-NIOSH Heat Safety Tool app and WBGT monitor to assess heat stress risk in agriculture. Journal of Occupational and Environmental Hygiene 2020;17(4):181–192. https://stacks.cdc.gov/view/cdc/230582
4. MoHRE announces return of the Occupational Heat Stress Prevention Policy, 15 June to 15 September 2026 (22nd year). WAM, 8 June 2026. https://en.aletihad.ae/news/uae/4671066
5. Midday Break 2024 with 6,000 delivery rest stations and an interactive map. https://en.aletihad.ae/news/business/4498519 . The 10,000+ stations figure for 2025 is from the MoHRE announcement in your earlier research notes; add its link before submitting.
6. Kenzen Launches Body Heat Sensor System for Worker Safety. T&D World. https://www.tdworld.com/electric-utility-operations/article/21131444/kenzen-launches-body-heat-sensor-system-for-worker-safety
7. EGA extends use of wearable technology to Beat the Heat in UAE summer. WAM. https://www.wam.ae/en/details/1395303185388 ; EGA encourages the use of wearable technology to detect body heat, Al Circle, 9 August 2023. https://www.alcircle.com/news/ega-encourages-the-use-of-wearable-technology-to-detect-body-heat-in-uaes-scorching-summer-97836
8. OSHA-NIOSH Heat Safety Tool App. NIOSH/CDC. https://www.cdc.gov/niosh/topics/heatstress/heatapp.html
9. Dubai Municipality distributes cool collars to labourers. Gulf Business, 30 August 2017. https://gulfbusiness.com/en/2017/middle-east/dubai-municipality-distributes-cool-collars-labourers ; NIA AC safety helmet, Khaleej Times (Expo 2020 Dubai). https://www.khaleejtimes.com/expo/video-worlds-first-ac-safety-helmet-launched-at-expo-2020-dubai
10. Stull R. Wet-bulb temperature from relative humidity and air temperature. Journal of Applied Meteorology and Climatology 2011;50:2267–2269.
11. ISO 7243 and ACGIH WBGT guidance; Dubai Municipality heat-stress guidelines. Use these to replace the illustrative thresholds in section 3 after you have read the official values.
12. Future Health Digital Innovation Challenge 2026, Climate-Resilient Worker Health challenge statement. https://hack2skill.com/event/futurehealthchallenge/